

## Section 2.1 Worksheet: Operations with Matrices

### Problem 1: size and entries

For

$$A = \begin{bmatrix} 9 & 20 & 21 \\ 1 & 5 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix},$$

find the sizes of  $A$  and  $B$ ,  $a_{23}$ , and  $b_{31}$ .

### Problem 2: equality

Find  $p$  and  $q$  so that

$$\begin{bmatrix} p & 9 \\ 20 & q \end{bmatrix} = \begin{bmatrix} 1 & 9 \\ 20 & 21 \end{bmatrix}.$$

### Problem 3: arithmetic

For

$$C = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 2 \\ -1 & 6 \end{bmatrix},$$

find  $C + D$ ,  $C - D$ , and  $3C - 2D$ .

**Problem 4: dimensions**

Determine whether each product is defined and state its size:

(a)  $(3 \times 4)(4 \times 2)$

(b)  $(2 \times 3)(2 \times 2)$

(c)  $(1 \times 5)(5 \times 1)$

**Problem 5: multiplication**

Compute

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 3 & 2 \end{bmatrix}.$$

Show all four row-column products.

**Problem 6: matrix equation and columns**

Write

$$\begin{cases} x + 2y - z = 9, \\ 2x + y + 3z = 20 \end{cases}$$

as  $A\mathbf{x} = \mathbf{b}$  and express  $\mathbf{b}$  as a linear combination of the columns of  $A$ .

## Section 2.2 Worksheet: Properties of Matrix Operations

### Problem 1

Solve  $4X - A = B$  for  $X$ , where

$$A = \begin{bmatrix} 4 & 8 \\ 12 & 16 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & 0 \\ 4 & 20 \end{bmatrix}.$$

### Problem 2

Verify  $3(A + B) = 3A + 3B$  for

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & -1 \\ 2 & 0 \end{bmatrix}.$$

### Problem 3

Compute  $AB$  and  $BA$  for

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}.$$

**Problem 4**

Verify associativity for

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}.$$

Compute both  $(AB)C$  and  $A(BC)$ .

**Problem 5**

Find  $M^T$  and  $(M^T)^T$  for

$$M = \begin{bmatrix} 1 & 9 & 20 \\ 21 & 0 & 5 \end{bmatrix}.$$

**Problem 6**

Verify  $(AB)^T = B^T A^T$  for the matrices in Problem 3.

## Section 2.3 Worksheet: The Inverse of a Matrix

### Problem 1

Find and verify the inverse of

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}.$$

### Problem 2

Determine whether

$$B = \begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix}$$

is invertible. Explain.

### Problem 3

Find the inverse of

$$C = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

by Gauss-Jordan elimination. Show every row operation.

**Problem 4**

Solve

$$\begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

using an inverse.

**Problem 5**

For

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},$$

find  $(PQ)^{-1}$  directly and by using  $Q^{-1}P^{-1}$ .

## Section 2.4 Worksheet: Elementary Matrices and LU

### Problem 1

Construct the  $3 \times 3$  elementary matrix for each operation:

- (a)  $R_1 \leftrightarrow R_3$
- (b)  $R_2 \leftarrow -2R_2$
- (c)  $R_3 \leftarrow R_3 + 4R_1$

### Problem 2

Use an elementary matrix to perform  $R_3 \leftarrow R_3 - 4R_1$  on

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \\ 4 & 7 \end{bmatrix}.$$

### Problem 3

Factor

$$B = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$$

as a product of elementary matrices.

**Problem 4**

Find an LU-factorization of

$$C = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ -2 & 2 & 4 \end{bmatrix}.$$

Show all multipliers.

**Problem 5**

Use the factorization to solve

$$C\mathbf{x} = \begin{bmatrix} 4 \\ 11 \\ 14 \end{bmatrix}.$$

Show forward and back substitution.

## Section 2.5 Worksheet: Markov Chains

### Problem 1

Determine whether

$$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}, \quad B = \begin{bmatrix} 0.6 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

are column-stochastic.

### Problem 2

For

$$P = \begin{bmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix},$$

find  $\mathbf{x}_1$ ,  $\mathbf{x}_2$ , and  $\mathbf{x}_5$ .

### Problem 3

Find the steady state of the matrix in Problem 2.

**Problem 4**

For

$$P = \begin{bmatrix} 0.6 & 0.2 & 0.1 \\ 0.3 & 0.7 & 0.2 \\ 0.1 & 0.1 & 0.7 \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} 0.4 \\ 0.4 \\ 0.2 \end{bmatrix},$$

verify that  $P$  is stochastic and find  $\mathbf{x}_1$ .

**Problem 5**

For the absorbing chain

$$P = \begin{bmatrix} 0.4 & 0 & 0 \\ 0.35 & 1 & 0 \\ 0.25 & 0 & 1 \end{bmatrix},$$

identify the absorbing states and find the limiting absorption probabilities starting from State 1.

## Section 2.6 Worksheet: Applications

### Problem 1: cryptography

Use blank = 0,  $A = 1, \dots, Z = 26$  and

$$E = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

to encode **RNB MUSIC**. Pad with a blank if necessary.

### Problem 2: decoding

Find  $E^{-1}$  and decode the first two coded blocks from Problem 1.

**Problem 3: Leontief model**

For

$$D = \begin{bmatrix} 0.20 & 0.10 \\ 0.25 & 0.20 \end{bmatrix}, \quad \mathbf{e} = \begin{bmatrix} 70 \\ 55 \end{bmatrix},$$

solve  $(I - D)\mathbf{x} = \mathbf{e}$ .

**Problem 4: least squares**

Find the least squares line for

$$(1, 4), (2, 7), (3, 8), (4, 13).$$

Show  $X$ ,  $X^T X$ ,  $X^T \mathbf{y}$ , the normal equations, and SSE.